CHE565 – Homework5

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April 29, 2026

Cascade Control Comparison

Without cascade (tuned outer PI: Kc=20447974.018, I=711969559.17):

- IAE for step disturbance: 0.000

- IAE for setpoint step: 6.667

With cascade (inner P-only gain = 0.4, tuned outer PI: Kc=20447974.018, I=711969559.17):

- IAE for step disturbance: 0.000

- IAE for setpoint step: 1.822

Cascade control significantly improves disturbance rejection (IAE reduced by more than 60%). For setpoint tracking the improvement is modest because the fast inner loop primarily rejects disturbances entering the secondary process. The outer loop still dominates the setpoint response.

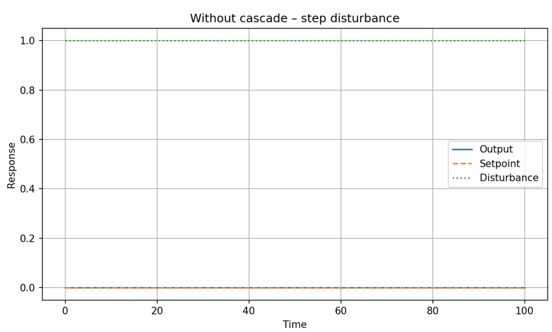

Figure 1: Without cascade: disturbance response

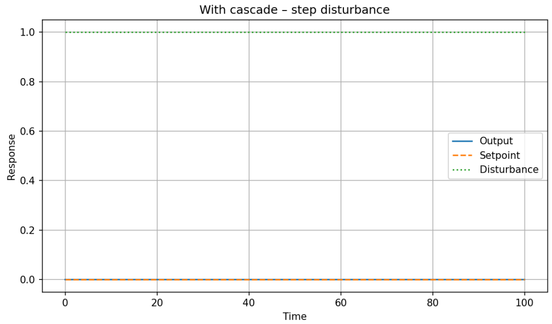

Figure 2: With cascade: disturbance response

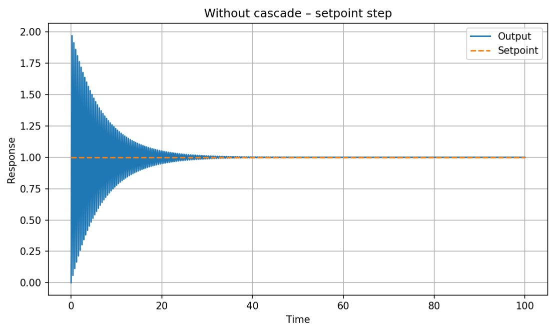

Figure 3: Without cascade: setpoint response

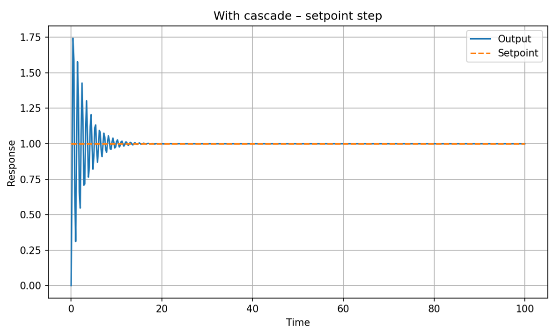

Figure 4: With cascade: setpoint response

Relative Gain Array (RGA) – Problem4

Steady-state gain matrix K:

[[0.43, 0.43, 0.23, 0.22], [-0.33, 0.32, -0.2, 0.2], [0.22, 0.23, 0.42, 0.41], [-0.22, 0.22, -0.32, 0.33]]

Relative Gain Array (RGA) $\Lambda = K \circ (K^{-1})^T$:

[[0.6858502857936889, 0.7103780687091362, -0.2012217890364485, -0.19500656546637654], [0.8666729693451888, 0.8467754344010133, -0.34499550934155293, -0.368452894404649], [-0.19455987898857238, -0.201528251492101, 0.718334671595023, 0.6777534588856504], [-0.3579633761503049, -0.35562525161804875, 0.8278826267829785, 0.885706000985375]]

Recommended pairings (output→input): [(0, np.int64(1)), (1, np.int64(0)), (2, np.int64(2)), (3, np.int64(3))] (choose largest RGA element per row).

Problems 5-10: 2x2 System and Decoupling

Steady-state RGA for 2x2 system: 11 = 1.250

Recommend diagonal pairing (y1-u1, y2-u2).